

# Physics-Guided Graph Learning for Conductive-Particle Allocation in Polydisperse Packed Beds

Research concept proposal

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## Abstract

Heat transfer through particle beds is important in many natural and industrial systems, including the heating, cooling, drying and temperature control of granular materials, as well as thermal-energy storage. Adding a small amount of highly conductive material may help, but its location can be as important as its amount. This study asks where conductive particles should be placed when the packing geometry, porosity, particle-size distribution and conductive-material volume must remain fixed. DEM will generate mechanically valid packings. A steady thermal-network solver will test candidate material allocations, while a repeated swap search will create strong reference solutions. A graph neural network (GNN) will then learn which particles are useful within the contact network. For a new packing, it will rank the particles that should receive the limited high-conductivity material. One final thermal solve will verify the proposed allocation. The study will test whether this fast graph-based rule can outperform random placement and simple physics-based rankings while avoiding most of the repeated search.

## 1 Motivation and research question

Heat travels through a packed bed along connected particle contacts. Conductive particles are therefore useful only when their locations support heat transport between the hot and cold boundaries. Random mixing may waste material in weakly used regions. Choosing only particles with many contacts may also fail because such particles can lie on side branches that carry little heat.

The central question is therefore:

*With the packing and amount of conductive material fixed, where should the conductive particles be placed, and can a GNN make this decision for a new bed?*

Figure 1 shows the task. DEM fixes the particle positions, sizes and contacts. We change only which particles are labelled as high conductivity. The study therefore measures the effect of placement alone.

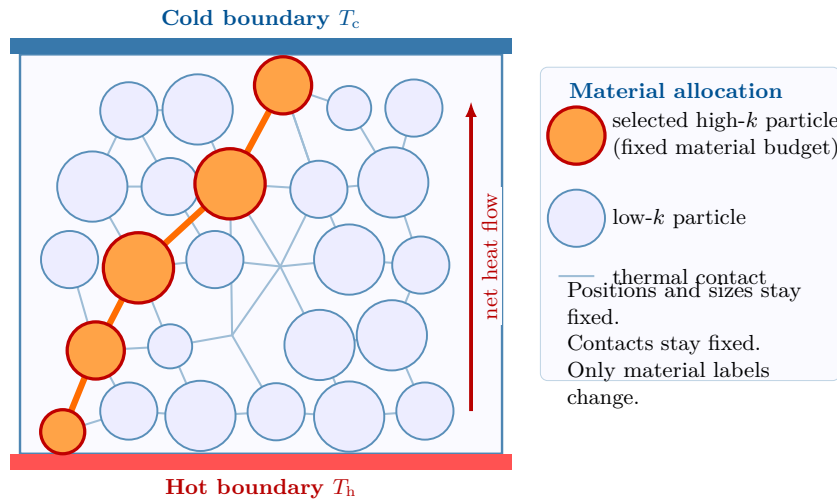


Figure 1: Inverse allocation problem in a polydisperse packed bed. The orange particles illustrate one possible conductive path; finding useful allocations is the subject of the proposed study, not an assumed result.

## 2 Hypothesis and objectives

The working hypothesis is:

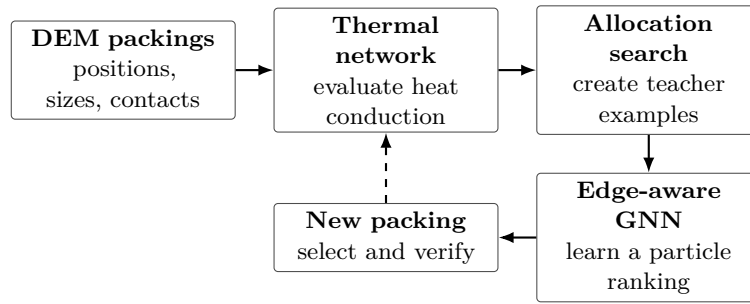
*Good allocations depend on connected paths through several particles. A GNN should recognize these paths better than rules that score each particle only from its own properties or number of contacts.*

This is a hypothesis to be tested, not an assumed result. The objectives are to:

1. measure how much particle placement matters at a fixed material budget;
2. create strong reference allocations using repeated thermal calculations;
3. test whether a GNN can propose good allocations for unseen beds; and
4. identify the contact-network structures that form useful heat paths.

## 3 Proposed hybrid modelling framework

Each component has one clear role:



**WP1: Controlled packing ensemble.** Generate independent polydisperse DEM beds with the same number of particles, size distribution and target porosity. The allowed number and size classes of high-conductivity particles remain the same in every case.

**WP2: Thermal solver and physics-guided search.** Solve steady heat conduction on the contact network and calculate the effective conductivity. Repeatedly exchange one selected and one unselected particle of the same size class. This fixed-budget swap search produces strong training examples while keeping particle count, size distribution and material volume unchanged.

**WP3: Graph-learning model.** Represent particles as graph nodes and their thermal contacts as edges. For a new packing, the GNN assigns one usefulness score to every particle. Particles are ranked separately within each size class, and the highest-ranked particles receive the available high-conductivity material. Thus, the GNN predicts *where to place the material*; it does not generate the packing itself.

**WP4: Transfer test and mechanism analysis.** Keep complete packings unseen during training. On each unseen bed, compare the GNN allocation with random placement, contact-number ranking, a simple heat-path ranking and the swap search. Verify every proposed allocation with the same thermal solver and inspect the resulting heat paths.

## 4 Expected contribution, scope and decision criterion

The expected contribution is a fast material-allocation method for disordered particle beds. The thermal solver determines which allocations work, the swap search creates reference examples, and the GNN makes this knowledge reusable for new beds. Unlike a model that only predicts effective conductivity, the GNN selects concrete particles, so its proposed paths can be inspected directly.

The first study will isolate stationary solid-contact conduction. Convection, radiation, particle motion and application-specific material constraints will remain outside the initial scope. They can be added if the controlled study shows that allocation is important.

The GNN is useful only if, on unseen packings, it consistently performs better than random placement and the simple rankings, recovers a meaningful part of the swap-search improvement, and needs far fewer thermal solves. If a simple rule performs equally well, the study will show that the GNN is unnecessary.